

Question ID: c5479c1a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A shipment consists of **5**-pound boxes and **10**-pound boxes with a total weight of **220** pounds. There are **13** **10**-pound boxes in the shipment. How many **5**-pound boxes are in the shipment?

Answer

- A. **5**
- B. **10**
- C. **13**
- D. **18**

Correct Answer: D

Rationale

Choice D is correct. It's given that the shipment consists of **5**-pound boxes and **10**-pound boxes with a total weight of **220** pounds. Let ***x*** represent the number of **5**-pound boxes and ***y*** represent the number of **10**-pound boxes in the shipment. Therefore, the equation **$5x + 10y = 220$** represents this situation. It's given that there are **13** **10**-pound boxes in the shipment. Substituting **13** for ***y*** in the equation **$5x + 10y = 220$** yields **$5x + 10(13) = 220$** , or **$5x + 130 = 220$** . Subtracting **130** from both sides of this equation yields **$5x = 90$** . Dividing both sides of this equation by **5** yields **18**. Thus, there are **18** **5**-pound boxes in the shipment.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of **10**-pound boxes in the shipment.